

# Lab exercises

## Phase 1: Discovery & Basics

| # | Exercise                                                                                                                                                                                                          | Example use case                                                                                                                                                                                                                                                                             | Cursor feature to highlight                            |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1 | <b>Understand the codebase. Start in Ask mode. There is no architecture diagram to begin from — have the model read the code and explain what the application does.</b>                                           |                                                                                                                                                                                                                                                                                              | Ask mode / codebase understanding                      |
| 2 | <b>Create a Canvas that shows both the technical architecture and the business flow.</b>                                                                                                                          | Example: Create a Canvas that visualize the business process and also the technical architecture of the project                                                                                                                                                                              | Canvas                                                 |
| 3 | <b>Plan / Implement. Use Plan and Implement to ship a new feature. Review the plan first and edit it before you start implementation if needed. After it lands, use the Bugbot subagent to review the change.</b> | Example: Add pagination to the incidents list (API + UI). Seed more demo data. Let Cursor write a script for you.                                                                                                                                                                            | Plan mode (/plan), built-in subagents (/review-bugbot) |
| 4 | <b>Write your first skill.</b>                                                                                                                                                                                    | Example: A skill that writes or updates documentation in a documentation.md file. Have the skill define the doc structure and how updates should be applied. Optionally, if you already have an MCP connection to Notion, Confluence, or similar, extend the skill to update docs there too. | Skills (/create-skill) (+ optional MCP)                |

## Phase 2: Modernization exercises

Practice using Cursor to modernize or refactor an app. Pick a path that fits you (or invent your own):

1. Migrate the project from React to Vue.js, or
2. Replace SQLite with a database you already know

| #  | Steps                                                                                                                                                                                                                                                                                                                                                                                | Cursor feature to highlight                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| M1 | Add a safety net. Before you migrate, use Cursor to add tests — frontend if you go React → Vue, backend if you swap databases. Solid coverage gives Cursor a feedback loop so it can check that behavior still matches after the change.                                                                                                                                             | Testing / safety net                             |
| M2 | Run the modernization. This repo is small enough to try a single “one-shot” prompt. For real migrations you’ll usually split the work across multiple agents (e.g. one finds what needs changing, another modernizes a component). Try different approaches and notice what works and what doesn’t. Steer with subagents, rules, skills, or Goals — there is no single right recipe. | Subagents / Multitasking / Goal / Skills / Rules |

## Phase 3: Add capabilities

Ship a larger product update to the app.

| #  | Exercise                                                                                                                                                                                                                                                              | Cursor feature to highlight                            |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| C1 | Internal admin dashboard. Build an admin page with usage / operational stats from the incident flow — e.g. how many incidents were created, closed vs still open, and main categories or root-cause themes. Stretch: self-service custom dashboards with saved views. | Subagents / Multitasking / Goal / Skills / Rules       |
| C2 | MCP for agent integration. Expose the QMS over MCP so external agents can plug in — e.g. Grok Bot can create or update an incident on the user's behalf, check status, or advance a step without desktop use. (App as MCP server.)                                    | Subagents / Multitasking / Goal / Skills / Rules / MCP |